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Concept Regions Matter: Benchmarking CLIP with a New Cluster-Importance Approach

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Abstract

Contrastive vision–language models (VLMs) such as CLIP achieve strong zero-shot recognition yet remain vulnerable to spurious correlations, particularly background over-reliance. We introduce Cluster-based Concept Importance (CCI), a novel interpretability method that uses CLIP's own patch embeddings to group spatial patches into semantically coherent clusters, masking them, and evaluating relative changes in model predictions. CCI sets a new state of the art on faithfulness benchmarks, surpassing prior methods by large margins; for example, it yields more than a twofold improvement on the deletion-AUC metric for MS COCO retrieval. We further propose that CCI when combined with GroundedSAM, automatically categorizes predictions as foreground or background-driven, providing a crucial diagnostic ability. Existing benchmarks such as CounterAnimals, however, rely solely on accuracy and implicitly attribute all performance degradation to background correlations. Our analysis shows this assumption to be incomplete, since many errors arise from viewpoint variation, scale shifts, and fine-grained object confusions. To disentangle these effects, we introduce COVAR, a benchmark that systematically varies object foregrounds and backgrounds. Leveraging CCI with COVAR, we present a comprehensive evaluation of eighteen CLIP variants, offering methodological advances and empirical evidence that chart a path toward more robust VLMs.

1. Introduction

Contrastive vision–language models (VLMs) such as CLIP [40] demonstrate strong generalization in tasks including zero-shot classification, retrieval [38], and open-vocabulary recognition [30, 31] across diverse domains [24, 34]. Despite this success, they remain vulnerable to spurious correlations [54, 57, 58], that is, associations driven by dataset biases rather than true semantic grounding [16]. For instance, in Figure 1(a), CLIP predicts water ouzel by relying on the surrounding water instead of the object. Such correlations reduce robustness under distribution shifts [9, 27, 53].

Recent work shows spurious correlations in VLMs often stem from dataset-specific artifacts, with object–background priors constituting a dominant source (e.g., zebras in grasslands, sharks in water) [36, 51, 53, 54, 59]. A notable attempt to quantify the role of background context is the CounterAnimals (CA) benchmark [54], which partitions images into "easy" and "hard" sets based on CLIP's accuracy, with the latter intended to probe sensitivity to atypical backgrounds. However, this operationalization is limited: declines in accuracy need not unambiguously signify reliance on background features, while elevated accuracy cannot be presumed to reflect object-centric reasoning.

Using our interpretability method CCI (Section 3), we analyze CLIP's (`ViT-B/16`) behavior on the CA dataset and show that accuracy-based partitioning is insufficient for diagnosing background sensitivity. As illustrated in Figure 1(a), correct predictions may rely on background cues (e.g., *water ouzel* identified via surrounding *water*), while errors can occur despite object-focused attention (e.g., *jaguar* misclassified as *cheetah*). Quantitative results reinforce this: although 32.1% of the easy set and 46.6% of the hard set are misclassified, the proportion attributable to background reliance remains nearly identical (6.23% vs. 7.26%, *BG-Er*, Figure 1(d)). Instead, most failures arise from fine-grained confusions (*Fine-Er*), indicating that raw accuracy is an unreliable proxy for disentangling object versus background-driven errors.

These findings highlight structural limitations of CA. By collapsing diverse visual variation into a binary easy–hard split, it overlooks heterogeneity in viewpoint, scale, pose, and composition. Errors therefore often reflect non-background factors; for example, failures on the *flamingo* class (Figure 1(b)) may stem from contrasting poses across

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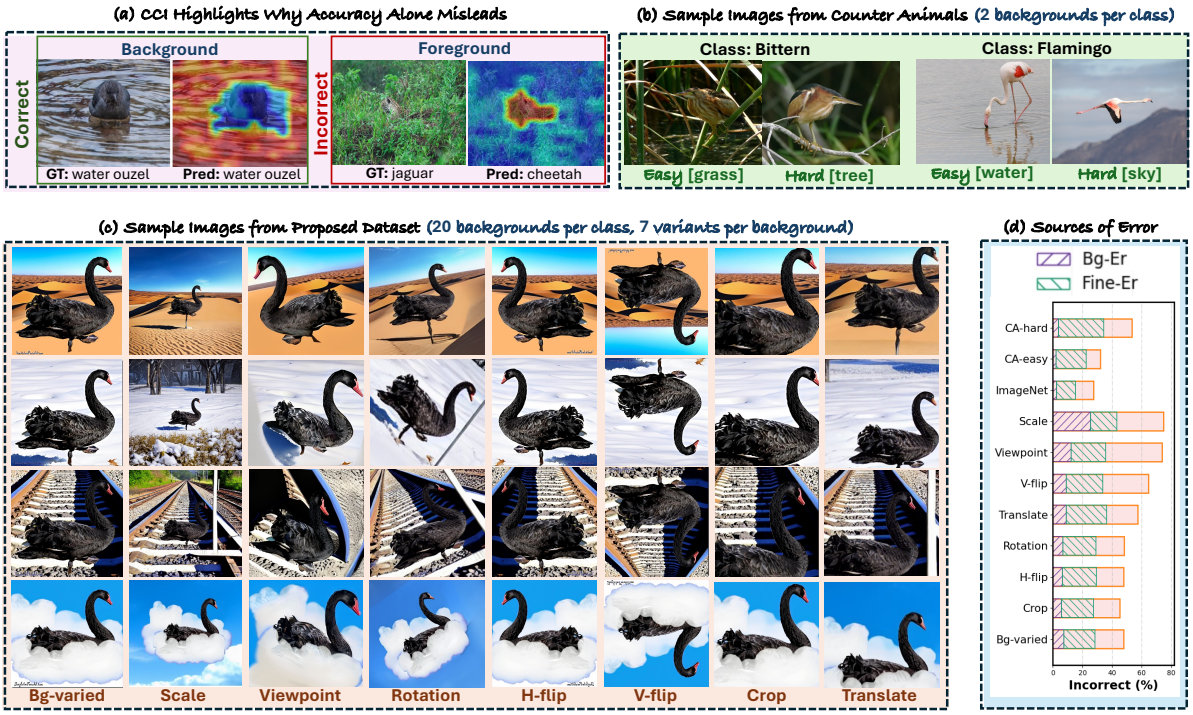


Figure 1. (a) Image and CCI maps for CLIP's prediction (red–blue heatmap, red = stronger attention). (b) Samples from easy and hard sets of CA. (c) Example of dataset curation in COVAR. (d) Proportion of different sources of errors in CA, ImageNet and COVAR subsets.

easy–hard splits rather than background context. As shown in Figure 1(d), background-driven errors (*BG-Er*) form only a small fraction overall. Consequently, CA's coarse partitioning and absence of fine-grained annotations restrict its diagnostic capacity and preclude precise assessment of CLIP's dependence on background correlations.

In summary, current benchmarks for spurious background correlations are deficient in two respects: (i) accuracy is an inadequate proxy, e.g., conflating background reliance with fine-grained class confusions, and (ii) effective evaluation requires controlled, diverse variation in visual factors to both isolate individual effects and expose a broader spectrum of failure modes.

To address these gaps, our first contribution is a training-free method, called Cluster-based Concept Importance (CCI), to interpret CLIP's predictions. CCI identifies the image regions driving image–text similarity by grouping patches into semantically coherent clusters, systematically masking them, and measuring changes in model predictions relative to the unmasked input (e.g. Figure 1, first row). In contrast to gradient-based approaches [61], which frequently yield noisy and fragmented pixel-level saliency maps, CCI identifies coherent and semantically meaningful regions, thereby enabling direct region-level interpretability (Figure 3). This property is crucial for our analysis, as CCI disentangles background and object contributions, providing a principled basis for diagnosing shortcut reliance and robustness failures in CLIP.

Our second contribution is COVAR (**CO**ntrolled **VAR**iants), a dataset developed for systematically evaluating models' reliance on spurious correlations. In contrast to the CA benchmark, which provides only two background variants and ignores other visual dimensions, COVAR introduces fine-grained and explicitly controlled variations. For each class, we begin with a base image (e.g., a *swan*, see Figure 1) and generate paired variants that independently perturb background, viewpoint, scale, translation, flip, rotation, and crop. Through this design, multiple factors shaping model behavior are explicitly represented and disentangled, thereby enabling rigorous, factor-wise evaluation.

As our third contribution, we benchmark 18 CLIP variants on CCI and our proposed dataset COVAR (Section 4) to assess their reliance on spurious correlations. We present results across eight subsets of COVAR, identify the dominant influences on model behavior, and provide insights for mitigating these vulnerabilities.

2. Related Works

Research on vision–language models spans *interpretability*, which explains predictive mechanisms, and *robustness evaluation*, which probes behavior under distribution shifts and challenging conditions. These are intertwined: inter-

pretability reveals the cues models rely on, while robustness tests whether such dependencies yield failures like spurious correlations.

Interpretability in Vision-Language Models: Early interpretability efforts applied methods such as attention rollout [1], which aggregates attention weights across layers, and gradient-based techniques like GradCAM [48] and GAME [5], which compute feature importance through gradients. With the advent of CLIP [40] and successors like BLIP [31] and ALBEF [30], researchers developed VLM-specific techniques: Grad-ECLIP [61] extends gradient-based attribution to multimodal settings, CLIP-Surgery [32] identifies important regions through similarity masking, MaskCLIP [63] performs token-level masking to localize influential regions, ECLIP [6] leverages gradient-weighted attention, and M2iB [55] employs perturbation-based explanations. Perturbation-driven methods such as RISE [39] further provide model-agnostic saliency through random masking. More recent concept-based approaches like SpLiCE [4] learn sparse concept embeddings, while others [25] use foundation models for concept discovery, building on earlier methods like Concept Bottleneck Models [28]. Yet both VLM-specific and traditional interpretability methods produce noisy, low-level attributions. Pixel-space methods such as SLIC [2], LIME [43], and SHAP [37] rely on superpixels that are not aligned with a model's representation. Perturbation-based approaches including RISE [39], Meaningful/Extremal Perturbations [12, 13], and the missingness analysis of [22] mask regions using a missingness token (e.g., black patches, noise, blur), making explanations sensitive to the replacement and often unstable. In contrast, CCI clusters CLIP's pretrained patch embeddings to obtain semantically coherent, representation-aligned regions and attenuate the model's attention to the corresponding patch tokens directly, avoiding pixel-level artifacts, preserving the input distribution, and yielding more faithful and localized attributions.

Benchmarks for Robustness and Spurious Correlations: Complementing interpretability work, robustness benchmarks have evolved from broad evaluations of adversarial robustness [17] and distribution shifts [18] to targeted assessments of specific weaknesses. ImageNet-A/R/C [19] and WILDS [29] highlighted vulnerabilities to natural adversarial examples and domain shifts, while synthetic datasets like ObjectNet [3] and 3D-Common [23] enabled systematic testing of pose and context. For VLMs, specialized benchmarks have emerged: Winoground [50] probes compositional reasoning, and Waterbirds [45] exposes background bias in bird classification. Most recently, CA [54] investigated CLIP's background sensitivity but, by relying on accuracy drops as a proxy. We extend this line of work by isolating visual factors and applying interpretability to distinguish spurious correlations from other errors.

3. Concept Cluster Importance

We present **Concept Cluster Importance (CCI)**, a training-free interpretability method for CLIP models that quantifies the contribution of semantically coherent visual concepts to image–text similarity scores. CCI operates entirely at inference time, requiring no model modification or retraining.

Patch Embedding Clustering. We focus on the patch embeddings $\mathbf{X} = \{\mathbf{z}_i\}_{i=1}^{N}$, which encode localized semantics. To extract coherent visual concepts, we perform K-means clustering over $\mathbf{X}$, yielding $\mathcal{C} = \{C_1, \ldots, C_K\}$, where each cluster aggregates semantically similar patches (See Figure 2 for examples with $K = 7$, where distinct colors denote different clusters). Ablations justifying choice of K are provided in Section A.1 of supp.

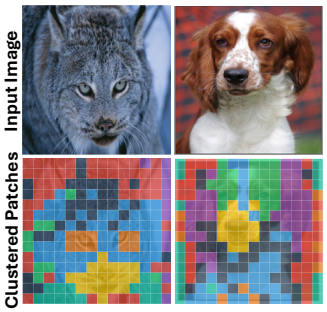


Figure 2. Patch clusters.

Preliminaries. Given an input image I, the CLIP image encoder (e.g., a Vision Transformer) processes I into a sequence of token embeddings $\mathbf{Z} = [\mathbf{z}_{\text{CLS}}, \mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_N] \in \mathbb{R}^{(N+1)\times d}$ where $\mathbf{z}_{\text{CLS}} \in \mathbb{R}^d$ is the global `[CLS]` embedding used for final image representation, and $\{\mathbf{z}_i\}_{i=1}^{N}$ are patch embeddings corresponding to N fixed-size patches extracted from the image.

Attention Attenuation with Cluster Masks. The CLIP image encoder contains L transformer layers, each with self-attention matrices $A^{(l)} \in \mathbb{R}^{(N+1)\times(N+1)}$, where the first token corresponds to the CLS embedding and the rest correspond to patches.

To measure the contribution of cluster C_k, we construct a binary mask $m_k(j)$ that indicates whether patch j belongs to cluster C_k:

$$m_k(j) = \begin{cases} 1, & j \in C_k, \\ 0, & \text{otherwise.} \end{cases}$$

The attention logits are then modified before softmax:

$$\hat{A}_k^{(l)}(i,j) = \begin{cases} A^{(l)}(i,j), & m_k(j) = 0, \\ -\infty, & m_k(j) = 1. \end{cases}$$

This masking is applied at every transformer layer and head, preventing the CLS token from aggregating information from the patches corresponding to the selected cluster.

Importance Scoring via Similarity Drop. Let $\mathbf{z}_{\text{CLS}}$ be the original final CLS embedding, and $\mathbf{t}$ the corresponding text embedding. The original similarity score is $s =$

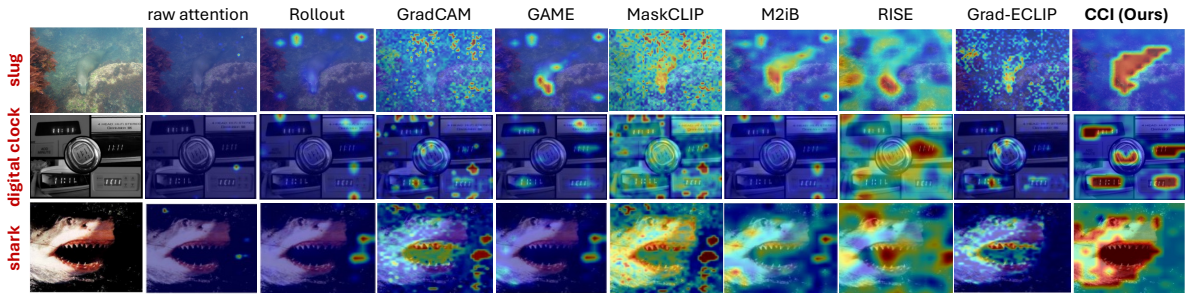


Figure 3. Qualitative comparison of CCI against baseline interpretability methods.

$\cos(\mathbf{z}_{\text{CLS}}, \mathbf{t}) = \frac{\mathbf{z}_{\text{CLS}}^\top \mathbf{t}}{\|\mathbf{z}_{\text{CLS}}\|\|\mathbf{t}\|}$. After attention attenuation for cluster k, we obtain a modified CLS embedding $\hat{\mathbf{z}}_{\text{CLS},k}$ and similarity $s_k = \cos(\hat{\mathbf{z}}_{\text{CLS},k}, \mathbf{t})$.

The relative importance of cluster C_k is quantified by the similarity drop $\Delta s_k = s - s_k$. We normalize the importance scores as $w_k = \frac{\Delta s_k}{\sum_{j=1}^{K} \Delta s_j}$, and compute the spatial importance map $S \in \mathbb{R}^{W \times H}$ as a weighted sum of cluster masks $S = \sum_{k=1}^{K} w_k \cdot m_k$. The map S identifies regions contributing most to the similarity score and is visualized as a heatmap overlay on I.

3.1. CCI Results

We evaluate CCI against a diverse set of baselines spanning gradient-based, attention-based, and perturbation-based techniques: Attention Rollout [1], GradCAM [48], GAME [5], MaskCLIP [63], M2iB [55], RISE [39], and Grad-ECLIP [61]. Unless otherwise specified, all experiments use CLIP with a `ViT-B/16` image encoder, and maps are computed with respect to the ground-truth label.

Qualitative Comparison with Baselines. Figure 3 compares attention maps from CCI and baseline methods. CCI consistently produces coherent, object-aligned heatmaps, whereas baselines yield sparse or noisy patterns. For instance, in the *slug* image (row 1), CCI captures the entire object while baselines highlight scattered regions; in the *digital clock* (row 2), CCI sharply localizes the digits on the clock face, relevant to CLIP's prediction, unlike baselines that miss or misfocus; and in the *shark* example (row 3), CCI emphasizes the teeth-features, whereas baselines diffuse attention across irrelevant regions. These results underscore CCI's key strengths: the clustering of semantically related patches into coherent concepts and the principled use of similarity-drop scoring to quantify their predictive contribution, together enabling precise, concept-level visualizations. Additional qualitative examples, results with other CLIP variants beyond `ViT-B/16`, and applications in the medical domain are provided in supp. (A.2, A.5 and A.6)

Quantitative Comparison. Consistent with prior work, we evaluate the faithfulness of CCI using deletion and insertion metrics [46]. CCI produces a patch-level importance map (14×14 for `ViT-B/16`), which is upsampled to 224×224 to assign pixel-level scores. Pixels are ranked by importance; in *deletion*, top-ranked pixels are iteratively replaced with random noise, while in *insertion*, they are progressively revealed from a blank canvas. At each step, ∼0.5% of pixels are modified, over 100 steps, cumulatively altering about half the image. The model's top-1 and top-5 accuracy is tracked at every step, and the area under the resulting curves (AUC) is used as a summary measure: lower AUC for deletion and higher AUC for insertion indicate causal influence of the highlighted regions. We report these scores on ImageNet-1K classification [10]) and MS COCO cross-modal retrieval [33].

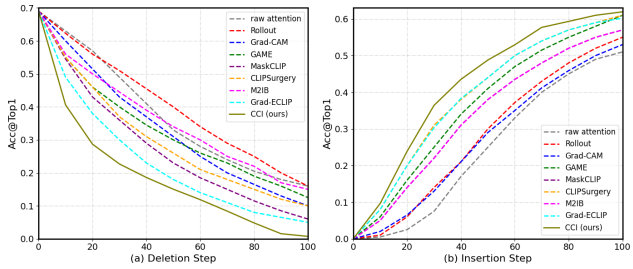


Figure 4. Deletion and insertion curves demonstrating CCI's quantitative superiority in identifying decision-relevant regions.

Across both datasets, CCI consistently outperforms all baselines. On ImageNet, deletion curves in Figure 4 drop sharply when removing regions identified by CCI, while insertion curves recover accuracy substantially faster, confirming that the highlighted regions capture the core evidence leveraged by CLIP. These trends are reflected in the AUC scores (Table 1), where CCI achieves state-of-the-art performance across Top-1 and Top-5 metrics. The same holds for image–text retrieval (Table 2), where CCI delivers state-of-the-art results for both image and text retrieval on COCO. Notably, in deletion, CCI attains over two-fold error reduction (0.2670 → 0.1056) in Top-5 IR, over the second-best Grad-ECLIP method. Collectively, these results show that CCI yields faithful, generalizable attributions of CLIP's decisions.

Table 1. Faithfulness evaluation of image explanations on *ImageNet* validation: AUC of Deletion/Insertion curves using Top-1 (@1) or Top-5 (@5) accuracy, with either ground-truth or predicted labels as CLIP text input.

Method	Deletion ↓				Insertion ↑			
	Ground-truth		Prediction		Ground-truth		Prediction	
	@1	@5	@1	@5	@1	@5	@1	@5
raw attention	0.3831	0.6239	-	-	0.2492	0.4195	-	-
Rollout	0.4082	0.6556	-	-	0.2803	0.4665	-	-
Grad-CAM	0.3417	0.5628	0.3518	0.5817	0.2682	0.4454	0.2526	0.4206
GAME	0.3356	0.5734	0.3497	0.5938	0.3611	0.5636	0.3425	0.5384
MaskCLIP	0.2848	0.4885	0.2886	0.4957	0.3335	0.5351	0.3275	0.5267
CLIPSurgery	0.3115	0.5235	0.3217	0.5412	0.3832	0.6021	0.3727	0.5719
M2IB	0.3630	0.5953	0.3633	0.5951	0.3351	0.5411	0.3347	0.5410
Grad-ECLIP w/o λ_i	0.2535	0.4379	0.2634	0.4568	0.3715	0.5831	0.3528	0.5556
Grad-ECLIP	0.2464	0.4272	0.2543	0.4420	0.3838	0.5993	0.3672	0.5749
CCI (Ours)	**0.1809**	**0.3276**	**0.1789**	**0.3318**	**0.4175**	**0.6518**	**0.3893**	**0.6201**

Table 2. Evaluation of **image** explanation faithfulness on *MS COCO* image-text retrieval (*Karpathy's split*) val-set: AUC for Deletion and Insertion curves for performance on image retrieval (IR) and text retrieval (TR) tasks.

Method	Deletion↓				Insertion↑			
	IR		TR		IR		TR	
	@1	@5	@1	@5	@1	@5	@1	@5
raw attention	0.1708	0.3554	0.1923	0.3720	0.1247	0.2552	0.1544	0.2969
Rollout	0.1948	0.3946	0.2268	0.4238	0.1294	0.2932	0.1753	0.3503
Grad-CAM	0.1717	0.3502	0.2161	0.4008	0.1027	0.2216	0.1152	0.2327
GAME	0.1706	0.3552	0.1982	0.3800	0.1537	0.3083	0.2097	0.3735
MaskCLIP	0.1321	0.2841	0.1516	0.2949	0.1423	0.2953	0.1891	0.3514
CLIPSurgery	0.1794	0.3652	0.2381	0.4292	0.1419	0.2941	0.1771	0.3384
M2IB	0.1797	0.3671	0.2057	0.3905	0.1469	0.3004	0.2058	0.3691
Grad-ECLIP w/o λ_i	0.1390	0.2940	0.1827	0.3386	0.1403	0.2895	0.1735	0.3279
Grad-ECLIP	0.1246	0.2670	0.1550	0.2933	0.1576	0.3203	0.2056	0.3761
CCI (Ours)	**0.0650**	**0.1056**	**0.0677**	**0.1184**	**0.1812**	**0.3513**	**0.2224**	**0.3943**

Understanding CLIP's Failure Modes: We use CCI to analyze CLIP's zero-shot predictions by visualizing attention maps with respect to the predicted class (Figure 5, ImageNet-1k in part a and CA in part b), focusing on misclassifications to expose error sources. Several failures are *foreground-driven*: in part a, row 1, a chimpanzee is misclassified as a siamang despite correct focus on the face; in row 2, attention is misdirected to a bucket, leading to error; and in row 3, CLIP fixates on a squirrel rather than the target class. In CA (part b, rows 2–3), CCI likewise reveals attention on the foreground, but errors arise from clutter, occlusion, or subtle visual distinctions.

Other errors are *background-driven*: in part a, row 4, attention to the grassy field results in *croquet ball*, while in part b, row 1, focus on background produces *water snake*. Even under occlusion or partial visibility (part b, rows 2 and 4), CCI consistently highlights the object, whereas baselines scatter attention across irrelevant regions. Overall, CCI reliably predicts CLIP's attention and offering clear interpretable insights, unmatched by existing methods.

Diagnosing Error Sources with CCI: While prior sections qualitatively showed that CLIP's errors extend beyond background bias to factors such as viewpoint, occlusion, and fine-grained confusion, here we use CCI to systematically measure the role of background cues. Using GroundedSAM [42], we obtain ground-truth foreground (FG) and background (BG) masks for ImageNet-1k and CA datasets (details are given in supp. A.7). CCI heatmaps are then computed with respect to CLIP's predicted class, and IoU overlap with FG/BG masks is used to classify predictions as *foreground-driven* (*FG-Er*) or *background-driven* (*BG-Er*).

For *foreground-driven* errors, we further identify cases of *fine-grained confusion* (*Fine-Er*). Using GPT-4o (prompt in supplementary), we assess whether the predicted and ground-truth categories are visually similar (e.g., *siamang* vs. *chimpanzee*) or unrelated, thereby distinguishing subtle fine-grained inter-class confusions from more severe errors caused by distractors or other factors.

Figure 1(d) summarizes errors: the first three rows show CA-hard, CA-easy, and ImageNet respectively. *BG-Er* constitute only a small fraction (9.1% on ImageNet, 6.7% on CA), with similar rates across CA's *easy* and *hard* sets, questioning the assumption that accuracy gaps primarily reflect background correlations. In contrast, a substantial portion of errors arise from *Fine-Er* (46.6% on ImageNet-1k, 60.4% on CA), underscoring the need to look beyond the background cues.

4. COVAR: A new benchmark

As discussed in Section 1 and shown in our CCI analyses, the CA benchmark is limited, offering only coarse background variation and no control over viewpoint, scale, flip, or crop. To address these gaps, we introduce COVAR, where each object is placed into multiple backgrounds and, for every such instance, its appearance is systematically varied along several visual factors.

4.1. Creating background variations

The objective is to synthesize diverse backgrounds for a given object. Unlike CA, which has limited diversity (45 animal classes), COVAR spans semantically and visually varied categories, from non-living objects (e.g., airships, locomotives) to living entities (e.g., birds, reptiles). A complete class list with ID mappings is provided in supp. A.8.2.

We select 33 classes from ImageNet and sample 50 images per class. Using the Emu2 image editing model [49], each image is synthesized across 20 curated background types (via GPT-4o [20] prompts; see supp. A.8.3) spanning outdoor (e.g., beach, railway track) and indoor (e.g., living room, kitchen) contexts, resulting in 1000 images per class. Figure 1(c) illustrates this process with a swan across four different backgrounds.

We evaluate the 33,000 background-varied (Bg-varied) images using CLIP `ViT-B/16` in zero-shot mode over the

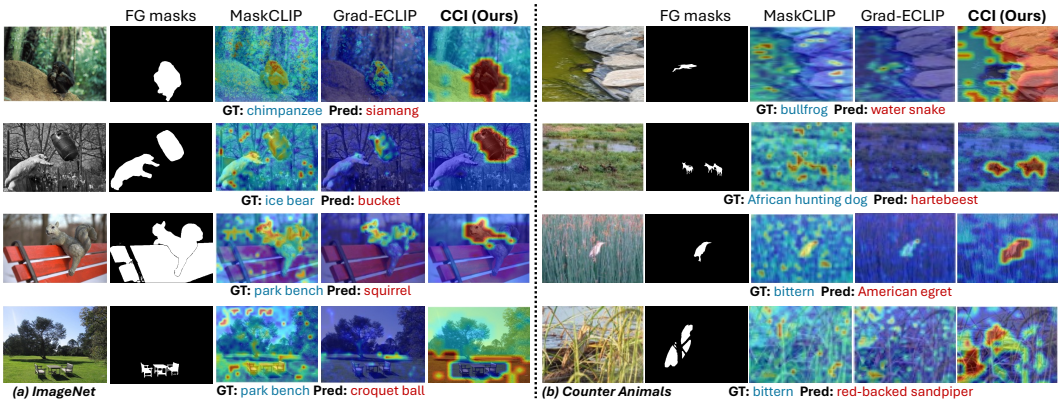


Figure 5. CCI analysis of CLIP failures on (a) ImageNet and (b) Counter Animals. Rows show the image, ground-truth foreground (FG) mask obtained using GroundedSAM, and attribution maps from MaskCLIP, Grad-ECLIP, and CCI, with ground-truth (GT) and predicted (Pred) labels below.

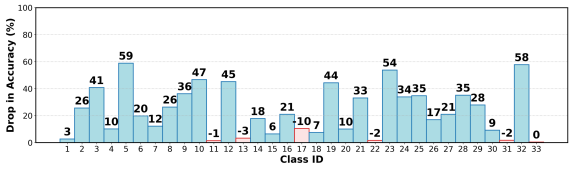


Figure 6. Class-wise accuracy drops for Bg-varied set.

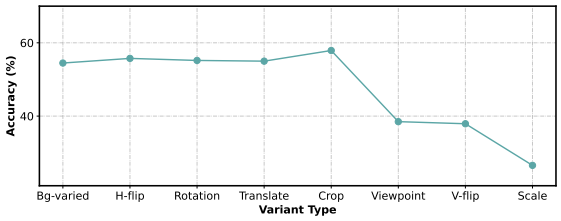


Figure 7. Subset-wise accuracy drops.

full 1000 class-labels from ImageNet. Figure 6 reports the average per-class accuracy drop relative to the original ImageNet images. While the mean drop is 23.78, the effect is highly uneven: classes 11, 22, and 31 show little-to-no decline, whereas classes 5, 23, and 32 suffer drops exceeding 50%, indicating that some categories are robust to background variation while others are strongly background-dependent. We show similar per-class accuracy drops for CLIP variants other than `ViT-B/16` in supp. A.9.

4.2. Extending with Structured Variants

To study additional factors influencing CLIP's accuracy, we extend each of the 33,000 Bg-varied images into 11 structured transformations: four different scales, two different viewpoints, horizontal and vertical flips, and single versions of translation, crop, and rotation (see supp. A.10

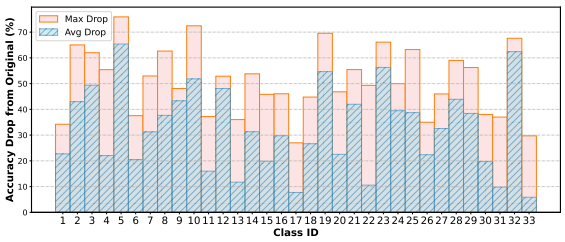


Figure 8. Class-wise average and maximum accuracy drops across variants.

for implementation details). Figure 1(c) illustrates this setup: the first column shows the object in varied backgrounds, while the remaining columns show transformed images. Together with the Bg-varied images, this expansion results in 396,000 samples, constituting the complete COVAR dataset.

We conduct zero-shot classification with CLIP `ViT-B/16` on eight COVAR subsets (bg-varied and seven transformations). Figure 7 shows that when compared with Bg-varied set, factors such as scale, v-flip and viewpoint cause steep declines. Hflip, rotation and translate lead to little or no degradation, crop leads to minor gains.

We evaluate class-wise accuracy drops (Figure 8), for each class, accuracies are computed across the eight subsets, averaged, and compared to the accuracy on the original ImageNet images. We also record the drop for the subset with maximum decline. The observed drops are substantially larger than those for background alone. Notably, even when average drops are modest, individual subsets can show severe degradation. For example, for class 22 the average drop is 10% while one subset exceeds 50%. Comprehensive subset-wise results for all classes are provided in the supplementary material (Section A.11).

Diagnosing error sources. We systematically analyze *BG-Er* and *Fine-Er* across all COVAR subsets, benchmarking against CA and original ImageNet (Figure 1(d)). The Bg-varied subset exhibits a notable rise in *BG-Er* relative to CA (6.7 to 15.6%). Scale and viewpoint variations further elevate both overall error rates and the proportion of *BG-Er* failures, with scale showing the strongest effect, indicating increased reliance on background cues under these transformations. In contrast, vertical flip, horizontal flip, translation, and crop do not exhibit such increases. Across all subsets, however, *Fine-Er* remains the dominant source of error.

5. Evaluation and Discussion

We benchmark CLIP variants from OpenCLIP [21] and OpenAI's original models [40] on COVAR, varying backbone size, patch resolution, and pretraining data (e.g., DataComp [15], LAION [47], DFN [11], WebLI [7]). We also evaluate SigLIP variants [52, 60]. Tables 3 and 4 present classification accuracy along with *BG-Er* and *Fine-Er*.

Overall performance: Among all models, `ViT-H-qgelu(DFN-5B)` at `378px` achieves the highest average accuracy of 56.8% (Table 3). `ViT-B-SigLIP2` performs strongly at `384px` and `512px`, while `ViT-SO-SigLIP2` maintains competitive accuracy even at `224px`, underscoring its training efficiency. Larger models such as `ViT-bigG` also show reasonable performance.

Performance across eight subsets: On the Bg-varied set, most models retain accuracies above 55%. Cropping does not hinder performance and often improves it, while H-flip and Translation cause only minor declines (e.g., large models such as `ViT-bigG` remains above 62%). Rotation produces minor drops, V-flip somewhat larger, and Viewpoint and Scale lead to the most severe declines. Across all perturbations, larger models consistently surpass smaller ones; for example, under Rotation, `ViT-H-qgelu(378px)` achieves 57.4% accuracy versus 30.1% for `ViT-B/32(DataComp-1B)`, underscoring the benefits of greater capacity and training scale.

Among all variants, Scale is the most challenging. `ViT-L/14(DataComp-1B)` drops from 62.2% on Bg-varied to 32.6% under Scale, while even the stronger `ViT-H-qgelu(378px)` declines from 65.2% to 39.3%. Viewpoint changes also yield substantial drops (e.g., `ViT-L/14` to 30.3%), though unlike Scale they do not coincide with major increases in *BG-Er* (Table 4). The first row in A.12 in the supplementary demonstrates how reduction in object scale may cause reliance on background.

Background reliance. Accuracy drops alone do not explain why models fail. Reducing Scale not only lowers accuracy but also nearly doubles *BG-Er* relative to the Bg-varied subset (Table 4), reaching 50.7% for `ViT-B/32(DataComp-1B)`. By contrast, under flips, rotations, or viewpoint changes, *BG-Er* remains stable (e.g., 12–17% for `ViT-L/14`). The findings imply that errors observed under these perturbations are more indicative of broad robustness deficits than spurious background correlations, underscoring the importance of disentangling these distinct failure modes.

Model size and data: Comparisons across architectures show that model size alone does not guarantee robustness. Large models such as `ViT-bigG` and `ViT-H/14(DFN-5B)` attain higher accuracy on the Bg-varied set (Table 3) yet exhibit substantial *BG-Er* under Scale (≈30–33%, Table 4). In contrast, models trained on curated data (e.g., DataComp-1B) display reduced background reliance, indicating that pretraining data quality shapes shortcut behavior as much as model size.

Backbone and resolution effects. With fixed training data, architecture matters: within DataComp-1B, `ViT-B/16` records lower *BG-Er* than `ViT-B/32` across nearly all perturbations (30.5% vs. 50.7% under scale, Table 4), suggesting finer patching helps reduce background reliance. Scaling up models (e.g., `ViT-bigG`) achieve higher raw accuracy (Table 3) but still sustain background-driven errors (32.7% in scale, Table 4). Similarly, higher input resolutions (e.g., `ViT-B-SigLIP2` at `512px`) improve Bg-varied accuracy (64.9%, Table 3) but only marginally reduce *BG-Er* (39.3% under scale, Table 4), showing that resolution alone does not mitigate background reliance.

Fine-grained confusion: Factors like clutter, occlusion, and viewpoint shifts hinder discrimination between fine-grained classes (e.g., *chimpanzee* vs. *siamang*). As shown in Table 4 (*Fine-Er* columns), such errors persist across subsets, underscoring the inherent difficulty of the COVAR dataset. At smaller scales (last column, Table 4), models trade off *Fine-Er* against *BG-Er*. At larger patch sizes (e.g., `ViT-B-SigLIP2/32`), reduced access to detail further obscures fine distinctions and instead amplifies reliance on spurious background correlations.

Takeaways. Our analysis highlights concrete avenues for enhancing robustness in CLIP-like models. Scale emerges as the most challenging perturbation, affecting both accuracy and background reliance, underscoring the need to explicitly address scale variation during training. One approach is multi-scale feature alignment [8], which fuses features across resolutions to reduce sensitivity to object size. Complementarily, RobustMixGen [26] augments data with diverse object–background combinations, enhancing robustness and mitigating reliance on spurious cues. Second, as viewpoint shifts degrade accuracy without increasing *BG-Er*, architectures that reduce over-reliance on 2D context, such as equivariant attention [44], may help. Third, the persistence of high *BG-Er* in large models in-

Table 3. CLIP variant accuracies across subsets of COVAR; final column shows overall average.

Name	Method			Bg-varied	H-flip	Translate	Crop	V-flip	Rotation	Viewpoint	Scale	Avg
	Patch	Res	Dataset	Acc	Acc	Acc	Acc	Acc	Acc	Acc	Acc	Acc
ViT-B	32	256	DataComp-1B	55.8	55.2	55.8	59.4	32.2	44.6	28.5	24.2	44.5
ViT-B	16	224	DataComp-1B	55.7	54.3	53.7	56.5	37.5	44.9	27.7	27.2	44.7
ViT-L	14	224	DataComp-1B	62.2	61.3	60.7	62.3	48.3	54.9	30.3	32.6	51.6
ViT-L	14	224	LAION-2B	59.7	58.9	60.0	60.2	43.2	53.9	32.2	30.1	49.8
ViT-H	14	224	LAION-2B	60.2	59.2	59.4	60.9	45.3	54.2	30.1	31.0	50.0
ViT-bigG	14	224	LAION-2B	61.6	62.1	62.2	63.8	45.7	56.5	34.2	33.5	52.5
ViT-L	14	224	DFN-2B	59.1	57.8	58.0	59.5	41.8	51.3	28.0	31.1	48.3
ViT-SO-SigLIP2	14	224	WebLI	63.3	62.7	62.4	63.6	53.0	58.6	34.1	36.9	54.3
ViT-B-SigLIP	16	256	WebLI	57.5	56.8	56.6	57.4	38.7	48.9	27.9	29.1	46.6
ViT-B-SigLIP2	16	384	WebLI	64.1	63.3	63.4	63.8	48.1	59.9	31.0	37.8	53.9
ViT-B-SigLIP2	32	256	WebLI	54.8	53.9	55.0	58.2	30.9	41.6	26.0	25.6	43.2
ViT-B-SigLIP2	16	512	WebLI	64.9	63.8	63.9	64.1	49.3	60.6	30.5	36.4	54.2
ViT-H-qgelu	14	378	DFN-5B	65.2	65.4	65.5	65.7	55.0	61.7	36.7	39.3	56.8
ViT-H-qgelu	14	224	DFN-5B	63.2	62.9	63.3	64.1	50.9	57.4	35.9	38.2	54.5
ViT-B	16	224	OpenAI	52.2	52.3	51.7	54.8	35.2	42.6	26.1	25.0	42.5
ViT-B	32	224	OpenAI	47.9	48.0	46.0	48.3	26.8	30.1	21.8	22.1	36.4
ViT-L	14	224	OpenAI	56.5	56.7	56.0	57.4	46.9	49.2	26.7	28.3	47.2
ViT-L	14	336	OpenAI	57.6	57.8	57.2	58.4	49.5	53.1	27.3	31.8	49.1

Table 4. *BG-Er* and *Fine-Er* across different subsets of COVAR.

Name	Method			Bg-varied		H-flip		Translate		Crop		V-flip		Rotation		Viewpoint		Scale	
	Patch	Res	Dataset	*BG-Er*	*Fine-Er*	*BG-Er*	*Fine-Er*	*BG-Er*	*Fine-Er*	*BG-Er*	*Fine-Er*	*BG-Er*	*Fine-Er*	*BG-Er*	*Fine-Er*	*BG-Er*	*Fine-Er*	*BG-Er*	*Fine-Er*
ViT-B	32	256	DComp-1B	23.6	40.9	21.2	44.6	18.9	46.2	18.5	47.7	22.6	33.3	21.3	39.5	22.9	29.5	50.7	16.1
ViT-B	16	224	DComp-1B	14.2	49.3	11.7	52.7	11.3	53.4	11.9	52.2	13.8	42.6	12.4	50.4	15.7	33.5	30.5	28.4
ViT-L	14	224	DComp-1B	15.7	51.8	13.5	56.0	12.4	56.6	12.2	56.6	15.2	50.2	14.5	53.9	17.1	36.0	30.2	31.2
ViT-L	14	224	LAION-2B	16.9	49.8	14.0	55.0	13.8	53.2	12.4	55.4	15.4	44.3	16.2	51.4	17.1	33.5	35.5	27.8
ViT-H	14	224	LAION-2B	16.4	54.8	13.6	59.4	13.0	58.7	15.1	56.3	15.9	49.6	14.8	55.2	18.2	35.1	33.8	30.9
ViT-bigG	14	224	LAION-2B	17.8	51.2	16.0	54.3	16.4	53.6	17.2	52.3	15.4	48.3	17.4	54.0	19.6	33.0	32.7	29.7
ViT-L	14	224	DFN-2B	15.7	50.2	13.7	53.2	14.2	54.1	15.5	52.3	15.2	45.6	13.5	54.4	17.4	34.7	29.6	31.8
ViT-SO-SigLIP2	14	224	WebLI	25.3	47.8	22.7	52.8	21.2	53.1	20.0	53.8	23.0	51.7	25.4	51.4	27.0	32.7	42.3	29.3
ViT-B-SigLIP	16	256	WebLI	16.5	49.2	13.5	53.6	13.0	53.8	13.0	54.4	15.2	44.3	14.4	52.6	17.1	33.8	33.6	26.7
ViT-B-SigLIP2	16	384	WebLI	19.8	51.5	17.5	55.2	15.9	56.7	16.6	55.2	19.7	49.9	20.2	54.5	25.4	33.1	36.8	28.3
ViT-B-SigLIP2	32	256	WebLI	36.0	33.6	35.8	35.0	37.2	34.6	31.2	42.7	36.1	24.5	32.3	24.7	34.7	20.8	45.7	7.7
ViT-B-SigLIP2	16	512	WebLI	20.3	51.8	18.1	55.2	16.7	55.8	17.2	53.1	20.3	50.1	20.2	53.8	24.8	31.9	39.3	22.3
ViT-H-qgelu	14	378	DFN-5B	15.9	54.4	13.4	57.8	13.4	57.8	12.7	58.5	14.9	54.4	15.4	57.1	16.8	36.3	30.4	33.1
ViT-H-qgelu	14	224	DFN-5B	16.2	51.3	13.8	55.7	13.6	54.8	14.3	54.2	15.6	49.2	15.0	52.5	17.4	34.7	29.4	23.2
ViT-B	16	224	OpenAI	15.6	43.7	13.8	48.2	14.0	48.8	13.1	48.0	14.2	37.9	16.0	45.1	16.8	31.3	33.9	23.7
ViT-B	32	224	OpenAI	14.0	37.3	12.0	41.5	13.7	42.0	12.6	46.3	13.2	30.2	15.3	34.4	14.5	26.4	38.2	21.4
ViT-L	14	224	OpenAI	17.1	44.3	14.9	48.2	15.2	48.1	16.8	45.2	15.6	44.6	16.8	46.0	16.7	34.3	30.1	12.1
ViT-L	14	336	OpenAI	15.3	46.4	12.8	50.8	13.3	49.7	11.7	50.9	13.4	48.1	15.3	49.2	14.4	35.4	28.4	30.4

dicates that scaling alone cannot mitigate spurious background reliance. Curated pretraining data that decouples objects from typical contexts is crucial, as exemplified by ObjectNet [3] and noise-robust training [56]. Together, these results underscore that data-centric strategies, rather than mere scaling, are essential for achieving improved robustness and generalization. Fourth, our backbone comparisons indicate that smaller patch sizes consistently mitigate background reliance. Hierarchical architectures such as the Swin Transformer [35], with finer-grained patch granularity, could support better object localization and reduced dependence on global context, making them a worthwhile direction for further investigation. Finally, persistent *Fine-Er* patterns indicate that robustness cannot be achieved by background debiasing alone. Prior work shows leveraging *fine-grained, category-specific textual descriptions* [41] and integrating *part-based localization* and *attention mechanisms* [14, 62] may help improve intra-class separability. Such strategies complement background robustness by helping models resist spurious correlations while capturing subtle distinctions needed for fine-grained categories. Overall, these findings point to the value of combining data curation, targeted augmentations, architectural refinements, and fine-grained supervision to advance model robustness.

6. Summary

We considered the problem of CLIP's vulnerability to spurious correlations during model prediction and proposed a new visual interpretability technique, called CCI, for analyzing and understanding these issues. CCI generated attention maps by identifying semantically relevant groups of pixels and evaluating model changes with these regions masked out in the input, resulting in state-of-the-art performance on standard faithfulness benchmark datasets. By combining CCI with GroundedSAM, we showed that existing benchmarks, such as CounterAnimals, are insufficient for properly characterizing CLIP's error behavior. Our results showed that while these benchmarks attributed performance degradation primarily to background, the true underlying reasons point to other visual factors such as viewpoint variations, scale shifts, and fine-grained confusions. Building on this observation, we next proposed a new benchmark dataset, called COVAR, which systematically synthesized all these variations for each object category. We then used COVAR to benchmark the performance 18 different CLIP variants to both summarize the current state of their performance and provide insights and clear recommendations for how to improve these models.

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